Section A — Concept Application

Question 1: Selecting CPU, RAM and Storage

Scenario: Nexus Innovations is purchasing new computers for employees who use programming software, web browsers, and video conferencing applications at the same time.

Answer:

When selecting hardware for Nexus Innovations, the CPU, RAM, and storage should be chosen according to the workload and compatibility requirements.

1. CPU (Central Processing Unit)

- A multi-core processor should be selected to handle multiple applications simultaneously.
- Clock speed should also be considered because it affects how quickly the processor performs tasks.
- CPU compatibility with the motherboard, including socket type and supported chipset, must be checked.
- Power consumption and cooling requirements should also be considered.

2. RAM (Random Access Memory)

- At least 16 GB RAM is recommended for programming, browsing, and video conferencing together.
- RAM generation, such as DDR4 or DDR5, must be supported by the motherboard.
- RAM speed, maximum capacity, and available memory slots should be checked.
- Compatible RAM modules help maintain stable system performance.

3. Storage

- An SSD should be preferred over an HDD because it provides faster boot times and application loading.
- An NVMe SSD can offer higher data transfer speeds than a SATA SSD.
- Sufficient storage capacity should be available for software, projects, and user files.
- The motherboard must support the selected storage interface.

Question 2: Antivirus Impact on System Performance

Scenario: After installing antivirus software on all startup office computers, employees notice that their systems have become slower.

Answer:

Antivirus software protects computers from viruses, malware, and other security threats. However, it can affect system performance because it uses CPU, RAM, and storage resources while running.

1. CPU Usage

Antivirus software uses CPU resources to scan files, monitor applications, and detect suspicious activities. Full system scans and real-time protection can increase CPU usage, making other applications run more slowly.

2. RAM Usage

Antivirus programs run background services and processes that consume RAM. If the computer has limited memory, fewer resources remain available for programming software, browsers, and other applications. This can cause slow performance and application delays.

3. Storage Performance

Antivirus software continuously reads and scans files stored on the disk. During full scans, it may perform many disk operations, increasing storage usage and causing delays when opening files or applications. This impact can be more noticeable on slower HDDs.

Conclusion: Antivirus software may slow down computers because it consumes CPU, RAM, and storage resources. To improve performance, the office should use lightweight antivirus software, schedule full scans during non-working hours, and ensure that computers have sufficient hardware resources.

Question 3: Troubleshooting Driver Incompatibility or Operating System Issues

Scenario: A USB keyboard and printer work properly on one computer but do not work correctly on another computer in the startup office.

Answer:

If a USB keyboard and printer work on one computer but not another, the problem may be caused by incompatible or outdated device drivers, USB port issues, or operating system problems.

The following troubleshooting steps should be performed:

1. Check Physical Connections

First, check whether the keyboard and printer are properly connected to the USB ports. Try connecting them to another USB port and check whether the devices are detected.

2. Check Device Manager

Open Device Manager in Windows and look for devices showing a yellow warning symbol or an unknown device message. These indicators may suggest a driver-related problem.

3. Update or Reinstall Drivers

Download the correct drivers from the official manufacturer website. Update the existing drivers or uninstall and reinstall them if they are corrupted or incompatible.

4. Check Operating System Compatibility

Verify whether the keyboard and printer support the installed operating system and its version. Install any required Windows updates if necessary.

5. Restart and Test the Devices

Restart the computer after installing or updating drivers. Reconnect the keyboard and printer and check whether they work correctly.

Question 4: UPS and Surge Protector

Scenario: Nexus Innovations experiences frequent power cuts and voltage fluctuations.

Answer:

Frequent power cuts and voltage fluctuations can damage computer components and cause unexpected system shutdowns. A UPS and surge protector help protect computer hardware and reduce the risk of data loss.

1. UPS (Uninterruptible Power Supply)

- A UPS provides temporary battery backup when the main power supply fails.
- It gives employees time to save their work and shut down computers safely.
- It helps protect systems against sudden power interruptions.
- Some UPS models also provide voltage regulation to handle certain voltage fluctuations.

2. Surge Protector

- A surge protector protects computer equipment from sudden voltage spikes.
- It helps prevent damage to sensitive components such as the motherboard, power supply, and storage devices.
- It works by limiting or diverting excess voltage away from connected equipment.

Conclusion: A UPS provides temporary backup power during outages, while a surge protector protects equipment from sudden voltage spikes. Using both helps improve hardware safety and reduce the risk of data loss.

Question 5: DNS and DHCP Working Together

Scenario: New employees at Nexus Innovations can reach internal systems, such as the application server, using IP addresses but not by name. Their devices sometimes fail to obtain a valid IP address when they connect to the network.

Answer:

DNS and DHCP are two important network services that help devices communicate within a network.

1. DNS (Domain Name System)

DNS converts human-readable domain or hostnames into IP addresses. For example, when an employee enters an application server's hostname, DNS finds the corresponding IP address so the computer can connect to the correct server.

2. DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns IP configuration to devices when they connect to a network. It provides information such as:

- IP address
- Subnet mask
- Default gateway
- DNS server address

3. How DNS and DHCP Work Together

When a new employee connects a computer to the office network, DHCP automatically provides the required IP configuration, including the DNS server address. The computer then uses DNS to resolve server names into IP addresses.

4. Most Likely Misconfigured Services

- If employees can access internal systems by IP address but not by name, the problem is most likely related to DNS configuration or name resolution.
- If devices fail to obtain valid IP addresses, the DHCP service may be unavailable, misconfigured, or unable to provide addresses from its available pool.

Question 6: Troubleshooting Firewall Rules

Scenario: After the IT team installs a new firewall at Nexus Innovations, employees can access the internet but are unable to reach the critical internal application server.

Answer:

A firewall monitors and controls incoming and outgoing network traffic based on configured security rules. In this situation, an incorrect firewall rule may be blocking legitimate traffic between employees and the internal application server.

The following troubleshooting steps should be performed:

1. Check Firewall Rules

Review the firewall configuration and examine rules related to the internal application server. Check whether the required source IP addresses, destination IP address, ports, and protocols are allowed.

2. Verify Server Details

Confirm that the application server is powered on, reachable on the network, and using the correct IP address. Also verify that the required application service is running.

3. Test Network Connectivity

Use the ping command to check basic connectivity to the server, if ICMP traffic is permitted. Use appropriate port connectivity tests to verify whether the required application port is accessible.

4. Check Firewall Logs

Examine firewall logs for blocked traffic. Look for denied connections between employee devices and the application server to identify which rule is blocking the communication.

5. Modify Firewall Rules

If a legitimate connection is being blocked, create or modify a firewall rule to allow the required traffic from authorized internal devices to the application server. Only the necessary ports and protocols should be permitted.

6. Test Again

After modifying the rule, test the application server connection from employee computers and confirm that the service is accessible.